MRI Segmentation via Training-Free Statistical-Graph Framework

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Abstract. Brain tumor subregion segmentation from multi-parametric MRI is essential for clinical triage, yet supervised deep learning is bottlenecked by scarce expert annotations, while classical unsupervised methods can be sensitive to initialization and have weak spatial constraints. To address these issues, we propose StaGra, a training-free unsupervised framework that integrates density-seeded statistical modeling with 3D spatial regularization. Following a preliminary tumor localization, StaGra initializes multi-parametric Gaussian Mixture Models at kernel density peaks for reproducible starting points, and subsequently minimizes a pairwise Markov Random Field energy using 3D α-expansion Graph Cut to enforce spatial coherence while preserving sharp anatomical boundaries. Evaluation on the BraTS 2021 benchmark yields Dice similarity coefficients of 78.4%, 49.2%, and 53.4%, and 95th-percentile Hausdorff distances of 15.2, 18.2, and 15.4 mm for whole tumor, tumor core, and enhancing tumor subregions, respectively, with GPU-free execution on standard CPUs. Overall, this work demonstrates a label-free approach at inference time, positioning StaGra as a practical tool for automated pseudo-label generation and coarse tumor localization in data-scarce environments.

Keywords: Glioma Segmentation · Unsupervised Learning · Adaptive Thresholding · KDE-GMM · 3D Graph Cut · Multimodal MRI · BraTS 2021

1 Introduction

Gliomas are heterogeneous brain malignancies comprising peritumoral edema (ED), necrotic core (NCR), and enhancing tumor (ET) [1,2]. Delineating these

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subregions from multi-parametric MRI (T1, T1ce, T2, FLAIR) is clinically vital but hindered by intensity overlap and irregular shapes [1]. Supervised deep learning is accurate but bottlenecked by costly, rater-dependent annotations [1], while unsupervised alternatives either ignore spatial context, fragmenting segmentations, or use local priors that over-smooth infiltrative boundaries [3,22]. We propose StaGra, an empirically calibrated unsupervised pipeline with no iterative weight optimization, label-free at inference; ground-truth labels serve solely offline, on a small BraTS 2021 calibration subset, for multi-objective threshold search. Our contributions are twofold:

1. **Density-Seeded Statistical Modeling:** We initialize GMM-EM [21] with deterministic kernel-density peaks [17] instead of heuristic seeding such as random initialization, K-means [16], or FCM [3], yielding a reproducible starting point; a paired ablation shows that the accuracy benefit is region-dependent and concentrated in the tumor core (Section 4.3).
2. **Volumetric 3D Graph Cut Spatial Regularization:** We minimize a pairwise MRF energy with α-expansion Graph Cut [4], extending prior 2D or supervised uses [18,19] to a fully unsupervised, volumetric setting in place of local mean-field updates [6]. Exact min-cut moves yield a strong local minimum within a constant factor of the optimum, while the contrast-sensitive pairwise term limits smoothing across intensity edges, preserving infiltrative borders with 3D coherence.

The remainder of this paper is organized as follows. Section 2 reviews related work, Section 3 presents the two-stage methodology, Section 4 reports experiments, and Section 5 concludes.

2 Related Work

Supervised architectures (e.g., 3D U-Net [8], Swin-UNet [5]) depend on massive annotated datasets, restricting clinical scalability [1], while classical clustering such as K-means [16] and Fuzzy C-means [3] avoids annotation but neglects spatial context and is vulnerable to bias fields and initialization volatility [1]. Statistical tissue classification via Gaussian Mixture Models (GMM) [21] offers a pretraining-free alternative, and because EM is susceptible to poor local minima under heuristic seeding, Kernel Density Estimation (KDE) [17] enables deterministic density-peak initialization [21]. Voxel-wise modeling nevertheless remains local and requires spatial regularization to suppress noise-driven misclassifications. MRFs [22] provide the framework for spatial priors, and methods differ in how the energy is optimized: local schemes such as mean-field updates [6] are prone to poor local minima, whereas α-expansion [4] solves a sequence of exact binary min-cut moves and converges to a strong local minimum within a constant factor of the global optimum for metric pairwise terms. Unlike hybrid methods relying on supervised classifiers for unary potentials [19] or restricted to 2D slices [18], StaGra is a fully unsupervised 3D pipeline integrating density-seeded multi-parametric GMMs with α-expansion Graph Cut minimization of a volumetric MRF energy.

3 Proposed Methodology

Our multi-class brain tumor segmentation framework operates in a probabilistic setting as an empirically calibrated unsupervised pipeline. Ground-truth annotations are utilized strictly offline for multi-objective hyperparameter search, rendering the framework completely label-free at inference time. As illustrated in Fig. 1, the pipeline consists of two sequential phases: Stage 1 localizes the candidate tumor region of interest (ROI), and Stage 2 performs density-seeded statistical mixture modeling with α-expansion Graph Cut regularization.

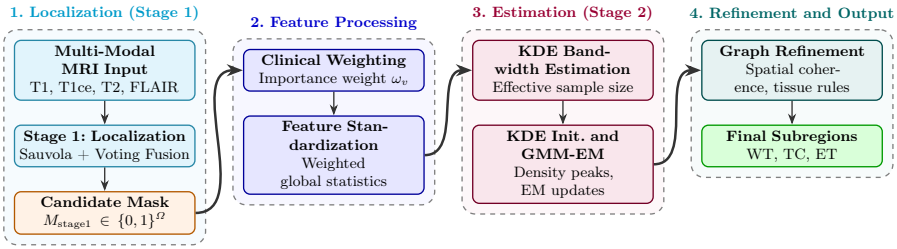


Fig. 1: The StaGra pipeline. A candidate mask M_{stage1} from multi-planar localization (Stage 1) restricts clinically weighted, standardized features, from which KDE density peaks seed GMM-EM (Stage 2); α-expansion Graph Cut then enforces 3D spatial coherence before latent clusters are mapped to WT, TC, and ET.

3.1 Problem Formulation

Let a multi-parametric 3D MRI study be defined on the voxel lattice $\Omega = \{1,\dots,H\} \times \{1,\dots,W\} \times \{1,\dots,Z\} \subset \mathbb{Z}^3$, where each voxel $v \in \Omega$ is observed through a normalized multi-sequence feature vector:

$$\mathbf{x}_v = [X_{v,\text{T1}}, X_{v,\text{T1ce}}, X_{v,\text{T2}}, X_{v,\text{FLAIR}}]^\top \in \mathbb{R}^4. \tag{1}$$

To eliminate the computational burden of processing healthy background tissues, statistical modeling is restricted to a candidate tumor ROI mask $M_{\text{stage1}} \in \{0,1\}^{\Omega}$ generated in Stage 1, defining the active voxel domain:

$$\mathcal{V} = \{v \in \Omega \mid M_{\text{stage1}}(v) = 1\}, \qquad N = |\mathcal{V}|. \tag{2}$$

Within domain $\mathcal{V}$, multi-class segmentation is formulated as finding an optimal latent label assignment $L^* = \{L_v\}_{v\in\mathcal{V}}$ over $K = 5$ statistical tissue classes (Necrotic Core – NCR, Peritumoral Edema – ED, Enhancing Tumor – ET, Cerebrospinal Fluid – CSF, and Healthy Brain Parenchyma – NL). This is achieved by maximizing the weighted mixture log-likelihood Θ^* under the Expectation-Maximization algorithm [21], followed by volumetric spatial regularization that minimizes the total MRF (Gibbs) energy [22,4]:

$$L^* = \arg\min_L \mathbf{E}(L \mid \Theta^*), \quad \mathbf{E}(L \mid \Theta^*) = \sum_{v\in\mathcal{V}} D_v(L_v) + \lambda \sum_{(u,v)\in\mathcal{N}} V_{u,v}(L_u, L_v) \tag{3}$$

where D_v and $V_{u,v}$ denote the unary and pairwise potentials, $\mathcal{N}$ is the set of adjacent voxel pairs under the 3D 6-connectivity neighborhood system, and $\lambda > 0$ controls the spatial smoothness trade-off.

3.2 Stage 1: Multi-Planar Localization

Inheriting the multi-planar adaptive thresholding pipeline of US-ATHC [1], Stage 1 extracts the active candidate mask M_{stage1} (Algorithm 1, Phase 1). To eliminate skull-stripped boundary artifacts at the brain border $\partial\Omega_{\text{brain}}$, we apply a **Median Background-Filling** [11] step prior to slice-wise binarization. The zero-valued background on 2D slice s is filled using the robust median intensity of brain parenchyma $\eta^{(s)} = \text{median}\left\{I(v) \mid v \in \Omega^{(s)}_{\text{brain}}\right\}$:

$$\tilde{s}(v) = s(v) \cdot \mathbb{1}\big(v \in \Omega^{(s)}_{\text{brain}}\big) + \eta^{(s)} \cdot \mathbb{1}\big(v \notin \Omega^{(s)}_{\text{brain}}\big). \tag{4}$$

Following intensity inversion $\bar{s} = 1 - \tilde{s}$, a candidate foreground voxel is admitted through the disjunction of three complementary rules: an *edge* rule from inverted adaptive Sauvola thresholding [15], a *global core* rule driven by a CV-adaptive coefficient k_g^*, and a *local core* rule based on per-slice statistics. The global coefficient is computed once per volume from the brain-wide coefficient of variation:

$$CV = \frac{\hat{\sigma}_{\text{brain}}}{\bar{\mu}_{\text{brain}}}, \qquad k_g^* = \min\left(k_{\max},\ \max\left(k_{\min},\ k_0 + \beta\,\frac{CV}{CV_{\text{ref}}}\right)\right), \tag{5}$$

while the local Sauvola threshold instead uses a fixed, empirically calibrated coefficient k_{Sauv} (Table 1), decoupled from k_g^* so that texture-scale sensitivity and volume-scale adaptivity are tuned independently. The bound $[k_{\min}, k_{\max}]$ in Eq. (5) prevents k_g^* from drifting to extremes under atypical brain-wide CV, i.e., under-segmentation on low-contrast volumes or false-positive healthy tissue on high-contrast ones. The three per-slice masks are combined as

$$M_{\text{edge}}(v) = \mathbb{1}\Big[\bar{s}(v) < \mu_w(v)\big(1 + k_{\text{Sauv}}(\sigma_w(v)/R - 1)\big)\Big] \wedge \mathbb{I}[s(v) > 0], \tag{6}$$

$$M_{\text{core}}(v) = \mathbb{1}\big[s(v) > \bar{\mu}_{\text{brain}} + k_g^*\hat{\sigma}_{\text{brain}}\big] \wedge \mathbb{1}[s(v) > 0], \tag{7}$$

$$M^{\text{local}}_{\text{core}}(v) = \mathbb{1}\big[s(v) > \bar{\mu}_{\text{slice}} + \kappa_{\text{loc}}\hat{\sigma}_{\text{slice}}\big] \wedge \mathbb{1}[s(v) > 0], \tag{8}$$

followed by removal of small connected components, morphological opening/closing, and hole filling (Algorithm 1, Phase 1). The three orthogonal orientation masks are then merged by weighted majority voting,

$$\text{vote}(v) = \alpha_a M_{\text{axial}}(v) + \alpha_c M_{\text{coronal}}(v) + \alpha_s M_{\text{sagittal}}(v), \tag{9}$$

$$M_{\text{voted}}(v) = \mathbb{1}\big[\text{vote}(v) \geq \tau\big]. \tag{10}$$

gated by post-fusion FLAIR intensity thresholding $\theta_{\text{gate}} = \bar{\mu}_{\text{brain}} + \kappa_{\text{gate}}\hat{\sigma}_{\text{brain}}$, and consolidated via 3D connected-component filtering and hole filling to yield the active candidate domain $\mathcal{V}$. All Stage 1 hyperparameters are summarized in Table 1.

Algorithm 1: StaGra: Two-Phase Tumor Segmentation

Input: Raw 4-channel MRI volume $\tilde{X} \in \mathbb{R}^{4\times H\times W\times Z}_{\geq 0}$, channels $\{T1, T1ce, T2, FLAIR\}$.

Parameters: Localization thresholds and fusion weights (Table 1); mixture order $K = 5$, EM tolerance δ, bandwidth range $[h_{\min}, h_{\max}]$, peak separation $d_{\min}$, seed boost b, Graph-Cut smoothness λ, cycles C.

Output: Nested tumor masks $\hat{Y}_{ET} \subseteq \hat{Y}_{TC} \subseteq \hat{Y}_{WT}$.

`// Phase 1 -- Multi-Planar Localization`

Normalize intensities and extract brain mask Ω_{brain}

for *each orientation* $o \in \{$*axial, coronal, sagittal*$\}$ **do**

  for *each 2D FLAIR slice* s *along* o **do**

    Fill background with slice median, invert intensities (Eq. (4))

    Compute local, global, and per-slice adaptive threshold masks $M_{\text{edge}}, M_{\text{core}}, M^{\text{local}}_{\text{core}}$ (Eq. (8), (5))

    $M_{\text{raw}} \leftarrow M_{\text{edge}} \vee M_{\text{core}} \vee M^{\text{local}}_{\text{core}}$; morphologically clean (remove small CCs, open, close, fill holes)

  Assemble per-orientation mask M_o from cleaned slices

Fuse $\{M_{\text{axial}}, M_{\text{coronal}}, M_{\text{sagittal}}\}$ by weighted majority voting (Eq. (10))

Apply post-fusion FLAIR intensity gate; consolidate via 3D connected-component filtering and hole filling

$M_{\text{stage1}} \leftarrow$ resulting candidate ROI mask

`// Phase 2 -- Density-Seeded Statistical Estimation` $\mathcal{V} \leftarrow \{v : M_{\text{stage1}}(v) = 1\}$

Compute clinical weights $\omega_v = \max\{\frac{1}{2}X_{v,\text{FLAIR}} + \frac{1}{2}X_{v,\text{T1ce}},\ 10^{-4}\}$, $\hat{\omega}_v = \omega_v / \sum_{u\in\mathcal{V}} \omega_u$; standardize features to $\mathbf{x}^{\text{std}}_v$

Estimate roughness-adaptive KDE bandwidth h^* per channel (Eq. (11))

Fit weighted KDE on a subsample $\mathcal{V}_{\text{sub}} \subset \mathcal{V}$; initialize $K = 5$ cluster means from the highest-density peaks with pairwise separation $\geq d_{\min}$ `// k-means (k-means++ init., fixed seed) fallback if fewer than` K `peaks`

repeat

  E-step: compute weighted responsibilities $\gamma_{v,k}$ (Eq. (12))

  M-step: update $\pi_k, \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k$

until $|\ell^{(t)} - \ell^{(t-1)}| \leq \delta(1 + |\ell^{(t-1)}|)$

Label the $K = 5$ clusters as $\{k_{\text{Healthy}}, k_{ET}, k_{CSF}, k_{ED}, k_{NCR}\}$ by tissue-intensity contrast (Eq. (15))

$\log p_{v,k}$ += $b \cdot \mathbb{I}[v \in S_k]$ for intensity-rule seed sets S_{ET}, S_{NCR}, S_{ED}

Compute unary/pairwise potentials $D_v(k)$, $V_{u,w}(k, k')$ (Eq. (13), (14))

for $c = 1$ **to** C **do**

  for $\alpha = 1$ **to** K **do**

    Update label field $\hat{L}$ via α-expansion min-cut over D_v and $\lambda V_{u,w}$

Construct nested masks $\hat{Y}_{WT}, \hat{Y}_{TC}, \hat{Y}_{ET}$ from $\hat{L}$ with morphological cleanup (Eq. (16))

return $\hat{Y}_{WT}, \hat{Y}_{TC}, \hat{Y}_{ET}$

3.3 Stage 2: Statistical Estimation

Density-Seeded GMM Estimation. Operating strictly within the candidate region $\mathcal{V}$ (Algorithm 1, Phase 2), Stage 2 estimates Gaussian Mixture Model (GMM) parameters $\Theta = \{\pi_k, \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k\}_{k=1}^5$ with diagonal covariances $\boldsymbol{\Sigma}_k = \text{diag}(\sigma^2_{k,1}, \ldots, \sigma^2_{k,D})$, using clinical voxel importance weights $\hat{\omega}_v$ via the Expectation-Maximization algorithm (GMM-EM) [21], modeling the five tissue classes introduced in Section 3.1 (NCR, ED, ET, CSF, healthy parenchyma). Because pointwise GMM-EM is non-convex and highly sensitive to initialization, volatile starting points frequently trap the algorithm in poor local minima [21]. To establish reproducible seeding, $K = 5$ initial cluster means $\{\boldsymbol{\mu}_k^{(0)}\}_{k=1}^5$ are selected as the highest-density peaks of a weighted Kernel Density Estimator (KDE) [17], subject to a minimum pairwise separation $d_{\min}$ that prevents redundant seeds

from collapsing onto the same dominant mode. Density estimation is evaluated over an active subsample $\mathcal{V}_{\text{sub}} \subset \mathcal{V}$ of size $N_{\text{sub}} = \min(N, N_{\max})$ drawn with a fixed random seed, reducing pairwise kernel evaluations from $\mathcal{O}(N^2)$ to $\mathcal{O}(N_{\text{sub}}^2)$ while retaining adequate representation of minor tissue modes (e.g., small ET subregions).

Standard Scott or Silverman bandwidth estimators [17] are prone to over-smoothing or bandwidth collapse under high-frequency imaging noise. We therefore adopt a roughness-adaptive bandwidth combining Silverman's AMISE-optimal scaling [17] with Kish's effective sample size [13]: for each channel d,

$$n_{\text{eff}} = \Big(\sum_{v\in\mathcal{V}_{\text{sub}}} \hat{\omega}_v^2\Big)^{-1},\; r_d = \frac{\sqrt{\text{mean}_b\left(\tilde{c}_{b+1}-2\tilde{c}_b+\tilde{c}_{b-1}\right)^2}}{\text{std}_b(\tilde{c}_b)},\; h_d^* = \frac{\sigma_d\, n_{\text{eff}}^{-1/(4+D)}}{(1+\gamma r_d)^{1/(4+D)}} \quad (11)$$

where $D = 4$ denotes the number of MRI channels (T1, T1ce, T2, FLAIR), $\tilde{c}_b$ is the Gaussian-smoothed weighted histogram of the standardized channel d, and the final bandwidth is $h^* = \min\big(h_{\max}, \max\big(h_{\min}, \text{mean}_d\, h_d^*\big)\big)$. This roughness correction damps the bandwidth in noisy or multimodal channels while recovering the standard AMISE rate in smooth ones, yielding stable, noise-robust peak detection without per-channel bandwidth tuning. Given the KDE-seeded means, weighted EM alternates between computing responsibilities

$$r_{v,k} = \frac{\pi_k\, \mathcal{N}(\mathbf{x}_v^{std}; \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)}{\sum_{k'} \pi_{k'}\, \mathcal{N}(\mathbf{x}_v^{std}; \boldsymbol{\mu}_{k'}, \boldsymbol{\Sigma}_{k'})}, \qquad \gamma_{v,k} = \hat{\omega}_v\, r_{v,k}, \quad (12)$$

and updating $\{\pi_k, \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k\}$ in closed form, until the log-likelihood satisfies $|\ell^{(t)} - \ell^{(t-1)}| \le \delta(1 + |\ell^{(t-1)}|)$. Full hyperparameters are listed in Table 1.

3D Graph Cut Optimization of the MRF Energy and Clinical Label Mapping. To go beyond pointwise posterior assignment and boundary-blurring local mean-field updates [6], we minimize the MRF (Gibbs) energy in Eq. (3) over the discrete label field L with α-expansion:

Unary potential (D_v): cost of assigning label L_v to voxel v via the GMM-EM negative log-posterior, with voxel-wise minimum-shift normalization for numerical stability:

$$D_v(L_v) = -\ln P(L_v \mid \mathbf{x}_v, \boldsymbol{\Theta}^*) - \min_k D_v(k) \quad (13)$$

Pairwise potential ($V_{u,v}$): penalizes label discontinuities between adjacent voxels $(u, v) \in \mathcal{N}$ with a contrast-sensitive Potts model, scaled by $\sigma^2 = \sum_{d=1}^{D} \text{Var}_{v\in\mathcal{V}}(x_{v,d}^{std})$ (computed once per volume):

$$V_{u,v}(L_u, L_v) = e^{-\frac{\|\mathbf{x}_u - \mathbf{x}_v\|^2}{2\sigma^2}}\, \mathbb{1}(L_u \neq L_v) \quad (14)$$

Energy minimization is approximately solved via a 3D Graph Cut algorithm using the α-expansion move paradigm [4], which converges to a strong local minimum within a constant-factor approximation bound of the global optimum, with spatial parameter $\lambda = 2.0$. The $K = 5$ clusters are deterministically labeled by ranking on tissue-intensity contrast, in decreasing order of prevalence and

Table 1: Hyperparameter summary for reproducibility. Sources: A = NSGA-II search [10]; B = fixed *a priori*; C = anatomical prior; D = hand-set (empirical). Algorithmic parameters are calibrated by the search, morphological constraints are fixed from anatomical properties.

Symbol	Description	Value	Src.
	Stage 1 — Multi-Planar Localization		
$w,\ R$	Sauvola window, dynamic range	25, 2.15	A
$k_{\text{Sauv}},\ k_0$	Sauvola & global-core coeffs. (Eq. (5))	0.21; 1.01	A
$\beta,\ CV_{\text{ref}}$	Adaptive-k slope, reference CV	0.30, 0.40	B
$k_{\min}, k_{\max}$	Adaptive-k clipping bounds	0.5, 2.5	B
$\kappa_{\text{loc}}, \kappa_{\text{gate}}$	Local-core / intensity-gate coeffs.	0.66, 0.29	A
$\alpha_a, \alpha_c, \alpha_s$	Fusion weights	0.50, 0.25, 0.20	B
τ	Vote threshold (Eq. (10))	0.66	A
$S_{\min}, r_{\text{open}}, r_{\text{close}}$	Min. 2D size, morphology radii	60 px; 3, 2	A
$\rho, S^{3D}_{\min}$	3D relative CC threshold, floor	0.0049; 600 vox	A
K_{top}	Top-K 3D components kept	3	C
	Stage 2 — Density-Seeded GMM & Graph Cut		
K, δ, iter.	Mixture order, EM tolerance, iter. cap	5; 10^{-6}; 150	B
$d_{\min}, N_{\max}$	KDE peak separation, subsample cap	0.15; 5000	B
$\gamma, [h_{\min}, h_{\max}]$	Bandwidth roughness penalty, clip range	0.1; [0.15, 3.0]	D
b, λ, C	Seed boost, Graph-Cut weight, cycles	8.0; 2.0; 3	D

specificity. Let $\mathcal{R} = \{0, \ldots, K-1\} \setminus \{k_{\text{Healthy}}, k_{ET}, k_{CSF}\}$ denote the two clusters remaining after the first three assignments:

$$k_{\text{Healthy}} = \arg\max_k \sum_v \mathbb{1}\big[\arg\max_{k'} \log p_{v,k'} = k\big], \quad k_{ET} = \arg\max_{k \neq k_{\text{Healthy}}} \mu_k^{T1ce},$$
$$k_{CSF} = \arg\max_{k \notin \{k_{\text{Healthy}}, k_{ET}\}} (\mu_k^{T2} - \mu_k^{FLAIR}), \quad k_{ED} = \arg\max_{k \in \mathcal{R}} \mu_k^{FLAIR}, \quad k_{NCR} = \mathcal{R} \setminus \{k_{ED}\} \tag{15}$$

i.e., ET is identified by peak T1ce contrast enhancement, CSF by the largest positive T2−FLAIR gap, and of the two remaining clusters the one with higher FLAIR intensity is assigned ED, the other NCR. The two clusters excluded from this assignment (k_{CSF} and k_{Healthy}) are designated Non-Lesional (NL) and omitted from further processing. Following established BraTS benchmark evaluation conventions [2], the final binary masks satisfy the anatomical nesting property $\hat{Y}_{\text{ET}} \subseteq \hat{Y}_{\text{TC}} \subseteq \hat{Y}_{\text{WT}} \subseteq M_{\text{stage1}}$:

$$\hat{Y}_{\text{ET}}(v) = \mathbb{1}(L_v^* = k_{\text{ET}}), \qquad \hat{Y}_{\text{TC}}(v) = \mathbb{1}(L_v^* \in \{k_{\text{NCR}}, k_{\text{ET}}\}), \qquad \hat{Y}_{\text{WT}}(v) = \mathbb{1}(L_v^* \in \{k_{\text{NCR}}, k_{\text{ED}}, k_{\text{ET}}\}). \tag{16}$$

4 Experiments

4.1 Experimental Setup

Dataset. We evaluate StaGra's unsupervised segmentation against non-learning baselines on multi-parametric MRI subjects (co-registered T1, T1ce, T2, FLAIR; 1 mm isotropic) from the BraTS 2021 benchmark [2]. Performance is assessed on WT, TC, and ET, with ground truth used for metric computation and, offline, for calibration only. The 1,251 subjects were split into a disjoint calibration set of 100 randomly allocated subjects and 1,151 unseen test subjects, on which all benchmarks and ablations are reported. Calibration is not sharply tuned: on the calibration set, fANOVA [12] over 129 search trials attributes 70% of the

Stage 1 Dice variance to κ_{loc} and at most 10% to any other parameter, and one-at-a-time perturbations (±10%, ±20%; one step for integer parameters) of 18 Stage 1 parameters change the mean Dice of the candidate mask by less than 1 pp for 14 of them. The exceptions are κ_{loc} (−3.83 pp at −20%), r_{open} (−10.14 pp at r_{open}=2), and the fusion rule (τ, α_a), which mainly shifts the recall–precision operating point of the candidate ROI, with recall decreasing by up to 7.67 pp (for α_a). Recall bounds the final segmentation ($\hat{Y} \subseteq M_{\text{stage1}}$), whereas an overly permissive ROI admits healthy tissue into Stage 2 modeling; the selected configuration (recall 0.78, precision 0.75) balances the two.

Baselines and Evaluation Metrics. StaGra is evaluated against six unsupervised baselines spanning thresholding, clustering, and spatial regularization: Multi-Otsu [14], MiniBatch K-means [16], FCM [3], GMM-EM [21], GMM-MRF [22], and Spatial FCM [7] (the latter two adapted to 3D 6-connected neighborhoods). For a fair comparison, all baselines share the same intensity normalization and a score-based candidate ROI (weighted T1ce,T2,FLAIR score, 0.85-quantile threshold, left–right asymmetry selection) that differs from StaGra's Stage 1; they use uniform voxel weights and no seed boost. Performance is quantified per subregion by the Dice similarity coefficient (DSC) and 95th-percentile Hausdorff distance (HD95) [20]. Joint empty masks yield DSC = 1 and HD95 = 0, whereas a single empty mask is penalized with DSC = 0 and HD95 = 45 mm.

4.2 Comparison with Unsupervised and Learning-Based Methods

Table 2 reports results on the 1,151 unseen test subjects. Top baselines reach only 50.30% DSC for WT (Multi-Otsu) and 26.00% and 38.00% for TC and ET (MiniBatch K-means), with HD95 > 52 mm. StaGra achieves WT/TC/ET DSC of 78.40%, 49.20%, and 53.40% and HD95 of 15.20, 18.20, and 15.40 mm, yielding absolute DSC gains of +28.10, +23.20, and +15.40 pp over the best baseline in each region. As baselines differ in localization, weighting, initialization, and regularization, these gains reflect holistic pipeline impact rather than a single component. On CPU, StaGra requires a median of 16.0 s per volume (Stage 1: 11.0 s; Stage 2: 5.0 s). We further compare StaGra on a test sub-cohort (Table 3) with supervised networks (Swin UNET [5], 3D U-Net [8]) and semi-supervised anomaly detectors (MViT-AE, 3D ResNet-18 [9]). Supervised models set upper bounds (Swin UNET > 87% DSC across subregions), yet StaGra surpasses both anomaly detectors in WT delineation (DSC 79.46% vs. 53.36% and 56.08%; HD95 14.51 mm vs. > 30 mm) without requiring healthy-control pre-training.

4.3 Ablation Study

Density-seeded GMM-EM initialization. We conduct a paired ablation holding all other components fixed (Stage 1 localization, clinical weighting, α-expansion Graph Cut) except the GMM-EM seeding strategy, comparing K-means centroid initialization against KDE density-peak initialization on the identical 1,151-subject test cohort, and report per-subject paired deltas (Δ =

Table 2: Comparison with six unsupervised baselines on 1,151 unseen BraTS 2021 test subjects (validation cases excluded). Cells: mean/median (IQR). Empty predictions get HD95 = 45 mm. StaGra outperforms all baselines on all metrics (paired Wilcoxon, Holm-corrected $p < 0.001$).

Method	**DSC (%) ↑**			**HD95 (mm) ↓**		
	WT	TC	ET	WT	TC	ET
Multi-Otsu	50.3/60.7 (59.0)	20.1/15.7 (35.6)	20.0/2.0 (44.3)	53.9/53.7 (42.7)	61.0/60.7 (40.5)	54.8/54.1 (48.4)
MiniBatch K-means	46.8/55.6 (64.6)	26.0/22.4 (45.8)	38.0/30.8 (76.1)	53.8/53.3 (44.0)	61.3/61.7 (41.7)	52.8/53.2 (51.9)
FCM	46.5/54.1 (65.8)	23.1/14.9 (44.1)	31.6/9.9 (73.5)	54.0/54.1 (43.0)	61.4/61.5 (40.6)	54.1/55.2 (46.6)
GMM-EM	44.1/50.9 (59.5)	24.1/17.2 (43.4)	28.1/11.3 (58.3)	54.2/54.4 (43.6)	61.7/61.8 (40.3)	55.1/57.0 (46.8)
GMM-MRF	44.1/51.2 (59.3)	24.2/17.4 (43.8)	28.2/10.8 (58.5)	54.2/54.2 (43.7)	61.8/61.8 (40.1)	55.0/57.0 (47.6)
Spatial FCM (3D)	46.7/55.0 (65.8)	23.1/14.7 (44.1)	32.3/9.8 (75.4)	54.0/54.1 (42.7)	61.5/61.5 (40.2)	52.5/53.7 (52.2)
StaGra (ours)	**78.4/86.5 (16.2)**	**49.2/53.5 (48.8)**	**53.4/61.3 (46.9)**	**15.2/7.8 (13.4)**	**18.2/11.4 (15.0)**	**15.4/7.1 (16.8)**

Table 3: StaGra vs. learning-based methods on a 300-case test sub-cohort (trained/validated on the other 951). T: (1) unsupervised, (2) supervised, (3) semi-supervised anomaly detection; "–": not reported.

		DSC (%)↑			HD95 (mm)↓		
Method	T	WT	TC	ET	WT	TC	ET
Swin UNET	2	92.63	89.70	87.41	6.85	4.98	4.36
3D U-Net	2	90.25	85.30	83.80	8.25	6.14	4.89
MViT-AE	3	53.36	34.95	41.98	30.59	36.37	35.29
3D ResNet-18	3	56.08	–	–	36.23	–	–
StaGra (ours)	**1**	**79.46**	**50.75**	**55.89**	**14.51**	**17.77**	**14.72**

Table 4: GMM-EM seeding ablation (KDE vs. K-means) ($n = 1151$). **Bold**: favors KDE.

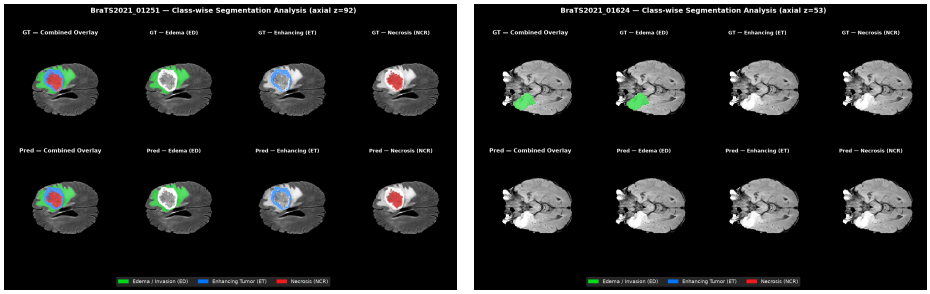
Region	Dice		HD95	
	Δ (pp)	Win (%)	Δ (mm)	Win (%)
WT	−0.74	**53.4**	2.82	40.2
TC	**2.13**	**54.4**	**−3.39**	**58.4**
ET	−3.58	21.7	3.19	22.4

Table 5: Spatial regularization ablation (1,151 unseen test subjects); cohort means.

	Dice (%)↑			HD95 (mm)↓		
Method	WT	TC	ET	WT	TC	ET
NoReg	73.70	43.92	51.29	31.97	35.95	30.40
MRF-MeanField	74.04	44.52	51.96	32.94	35.48	29.51
MRF-α-Exp (ours)	**78.40**	**49.20**	**53.40**	**15.20**	**18.20**	**15.40**

KDE − K-means) with KDE win rates. Table 4 shows a region-dependent effect rather than a uniform advantage: KDE seeding gives a modest gain for TC (Dice +2.13 pp, HD95 −3.39 mm; win rates 54.4% and 58.4%), WT is comparable (Dice win rate 53.4%, near chance), and K-means is better for ET on both metrics (KDE wins only 21.7% of subjects). KDE seeding thus mainly provides a reproducible starting point, while StaGra's gains stem primarily from Stage 1 localization and Graph Cut regularization (discussed next); as this ablation compares a single K-means run, robustness to random initialization was not measured.

Spatial Regularization: α-Expansion vs. Mean-Field. To isolate this contribution, we compare the full StaGra framework against unregularized assignment and local Mean-Field MRF smoothing, both operating on identical density-seeded GMM output and evaluated on the unseen test cohort of 1,151 subjects. Table 5 indicates that α-expansion minimization of the MRF energy outperforms both unregularized assignment and mean-field inference: compared



(a) BraTS2021_01251 (b) BraTS2021_01624

Fig. 2: Qualitative results on BraTS2021: a successful case (a) and a failure case (b), showing edema (ED), enhancing tumor (ET), and necrotic core (NCR) against the ground truth (GT).

to NoReg, MRF-α-Exp improves WT, TC, and ET DSC by 4.70, 5.28, and 2.11 percentage points and reduces HD95 by 16.77, 17.75, and 15.00 mm, respectively, whereas mean-field smoothing yields only marginal gains over NoReg (at most 0.67 DSC points). This suggests that, under our configuration, mean-field updates have limited effect on systematic misclassifications and fine infiltrative structures, whereas α-expansion promotes 3D volumetric spatial coherence while preserving sharp anatomical boundaries, as reflected in the lower HD95.

5 Conclusion

This paper presents StaGra, a training-free 3D brain tumor segmentation framework combining KDE-guided GMM-EM with α-expansion Graph Cut energy minimization. On 1,151 unseen BraTS 2021 subjects, StaGra outperforms six unsupervised baselines at a median CPU runtime of 16.0 s per volume. Limitations include candidate ROI dependence, offline calibration on a development subset, and single-dataset evaluation; future work will focus on uncertainty-guided localization and multi-center validation.

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